VECTOR

DaVinci Tools License Handling

Technical Reference

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Authors	Michael Hoffmann, Daniel Gamiz
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We have configured the programs in accordance with your specifications in the questionnaire. Whereas the programs do support other configurations than the one specified in your questionnaire, Vector´s release of the programs delivered to your company is expressly restricted to the configuration you have specified in the questionnaire.

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1 Introduction

The DaVinci Tools applications can be activated via the following kinds of licenses. The examples will be based on DaVinci Configurator, but they apply also for other DaVinci Tools.

For most our tools, you only need a Tool License to open the application, however, for a Tool like DaVinci Configurator, a SIP-based License is also needed.

> Tool licenses

This is the product license of the tool itself, for example: DaVinci Configurator PRO / Classic and all related variants. The available tool functionality depends on these licenses. The tool licenses are independent from the loaded DaVinci project (dpa file).

Following types of licenses are supported:

> Vector NVLM License

Legacy:

> Flexnet License – No longer supported after R25

> Legacy License Model (LLM) – No longer supported.



Note
For an overview on DaVinci Tools compatibility with the NVLM, please see following article: https://support.vector.com/kb/?id=kb_article_view&sysparm_article=KB0011956

> SIP-based Licenses

The SIP-based licenses depends on the currently loaded DaVinci project (dpa file). Some of the license flags in the SIP of the DaVinci project may enable tool features even without existence of a tool license.

2 Finding current used License Information

To find information of the current license being used by the tool, start the application and open the ‘Licenses’ dialog (Help > Licenses).This dialog will show the current License information.

In case of DaVinci Configurator, it provides two sections:

> SIP-License Details (‘Show SIP License Details’ button)

> Actual Tool License (‘Show Tool License Details’)

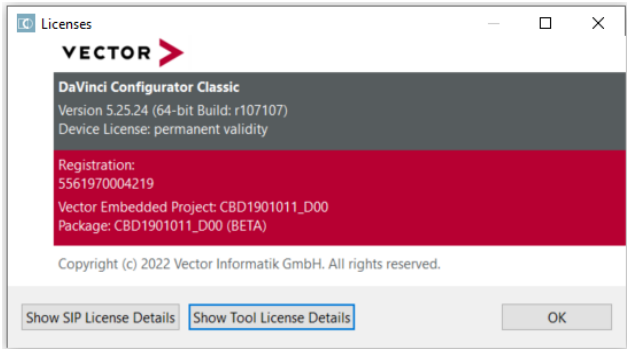


Figure 2-1 - Licenses Dialog

2.1 SIP-Based License

This section shows the currently available SIP-based license. If the DaVinci Configurator is being executed with a SIP-based license, the application main window header displayed “DaVinci Configurator SIP” + the SIP specific tool switches (.DIAG, .PB, …).

2.2 Tool Licenses

The following is a list of the current and legacy licenses that are available (server based or local) which could be used by the tool.

2.2.1 Current Licenses

- DaVinci Configurator Classic
- DaVinci Configurator Classic SE
- DaVinci Developer Classic
- DaVinci Developer Classic SE
- DaVinci Team

2.2.2 Legacy Licenses

- DaVinci Developer
  - Option .CPG
- DaVinci Configurator Pro

- Option .WF
- Option .MD
- Option .RTE
- Option .CI

License Display

Each entry within the sections of the license details dialog will have one of the following colors indicating whether the license is valid or not and used or not. Refer to following table for details.

License Appearance	
Color	**Description**
black	License is valid.
blue	License is valid and used by the current running application instance.
grey	License is expired or available for other (older) versions.

Table 2-1 License Appearance

Example below:

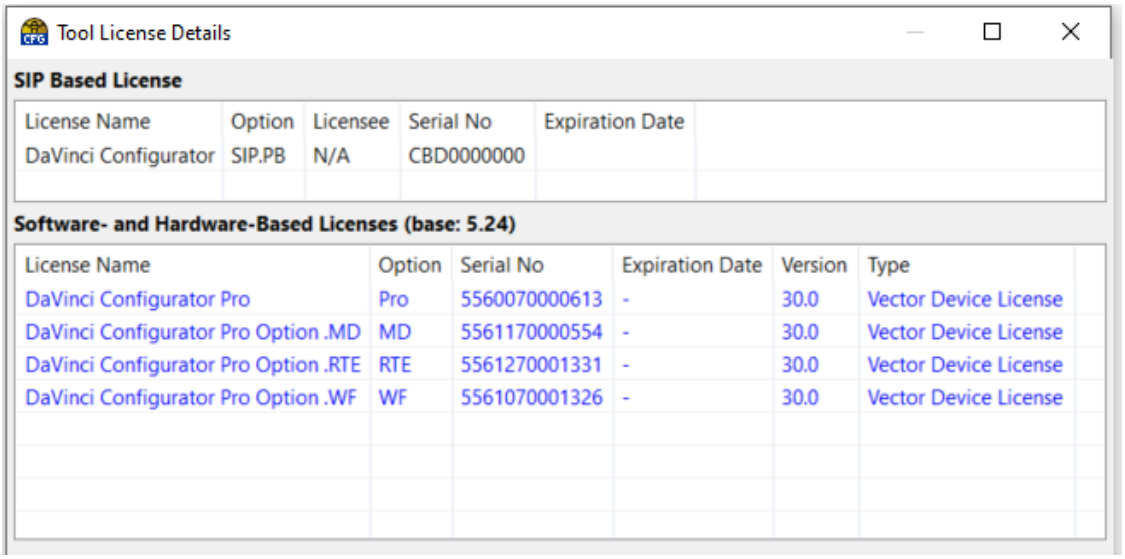


Table 2-2 Tool License Details: All Licenses are valid and currently used.

In case one of the licenses expires soon the expiration date column value shows a ‘(expires soon)’ tag within the licenses list.

2.3 Legacy: Display of License Server Configuration and Options

In case a FlexNet license server is configured (see 4.1.1 for details) the license details dialog shows the server configuration details and the according option configuration below the software- and hardware-based licenses list.

3 Vector NVLM Licenses

These are the different types of NVLM Licenses that currently exist:

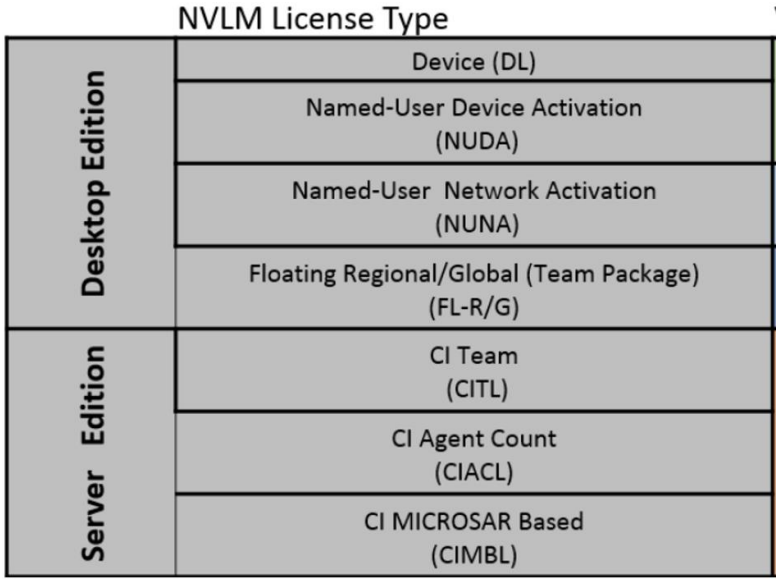


Figure 3-1 NVLM Licenses

The Desktop Edition licenses are meant to be used mostly with the Graphical User Interface (GUI), while the Server Edition licenses are meant to be used without the GUI.

The more common Vector NVLM License types are:

> Device License

> Named-User License

These are activated locally through the Activation Key on the Vector License Client.

The rest (NUNA, Floating and SE Licenses) can also be called Network-Based Licenses, as they required both a Server and a Client to work.

Reference
For further information on NVLM Licenses, please check the Vector KnowledgeBase on www.vector.com via Support & Downloads|KnowledgeBase|Get Answers|Vector License FAQs.

3.1 Network-Based Licenses

Network-Based Licenses require both a Server and a Client to work. These include:

> Floating Licenses

> Named User License (Network Activation)

> Server Edition Licenses: CI Team, CI Agent Count, CI MICROSAR Based

The On-Premise License Server (OPLS) is where the license is activated and the Client “borrows” the license from the Server whenever it is required.

3.1.1 On-Premise License Server (OPLS) Setup

As a Pre-requirement for the OPLS Setup, the following is required:

> Vector License Client (VLC) – Latest one, at least V6.2

  > The VLC can be downloaded through our website
  or

  > From can be found on the External Component Setup, also found on our website

Follow these steps to setup the OPLS:

1. Open, and set the VLC in Server Mode

> Done by clicking on Settings > Licensing Server > Enable License Server

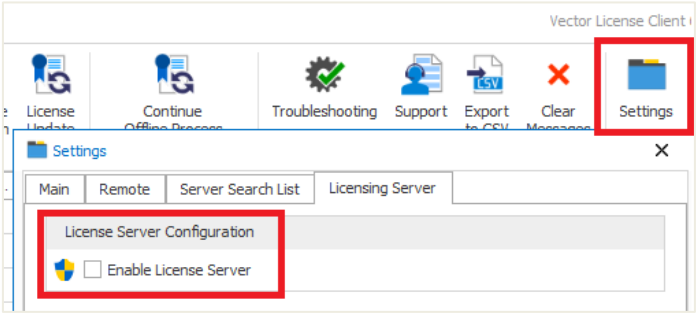


Figure 3-2 VLC License Server Configuration

2. With your Activation Key, activate your Network-Based License on the VLC
3. Use the command line `ipconfig`, to identify the OPLS IP Address (This is needed to set up the Client, as it needs to communicate with this IP)

```
Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . : vi.vector.int
   IPv4 Address. . . . . . . . . . . : 10.149.43.47
   Subnet Mask . . . . . . . . . . . : 255.255.255.192
   Default Gateway . . . . . . . . . : 10.149.43.1
```

Figure 3-3 Ip Address Example

3.1.1.1 Named User License Network Activation

In addition to the steps followed for OPLS above, you must specify the **Users** who should be able to reserve licenses. This is done through the following steps:

1. Access the **Manage User Assignment** button of the VLC.
2. Add the Usernames that you plan to assign to your license/licenses.

> Usernames are Case Sensitive

3. Add the Username to the right tool by clicking the checkbox
4. Transfer your changes to the server

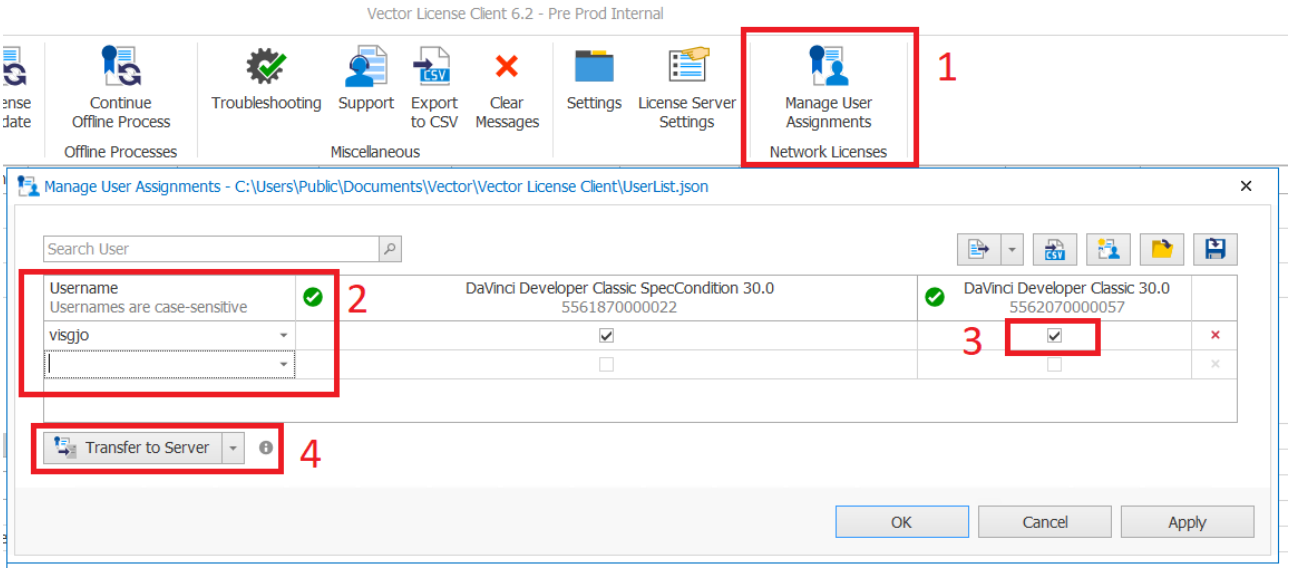


Figure 3-4 VLC User Assignments

3.1.2 Client Setup

3.1.2.1 Windows

1. Make sure you know the IP of your On-Premise License Server
2. Open the VLC
3. Add the OPLS IP Address to the search list (Screenshot example below),

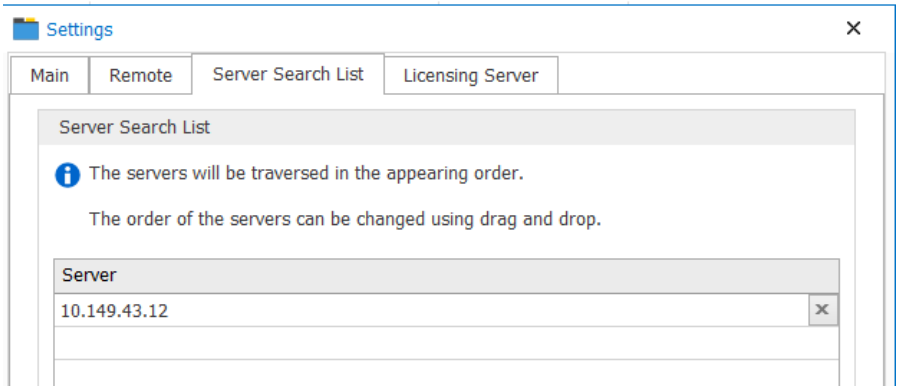


Figure 3-5 VLC Server Search List

4. The setup is complete. If the OPLS was correctly set up, you can now test the connection by opening a tool that needs one of the licenses set on the OPLS and check if the license is detected.

3.1.2.2 Linux

1. Make sure you know the IP of your On-Premise License Server,
2. Open the NVLM Server Search List site (Should work if you have VLC installed): http://localhost:22352/configuration/server_search_list.html
3. Add the IP to the search list (like shown in screenshot below)

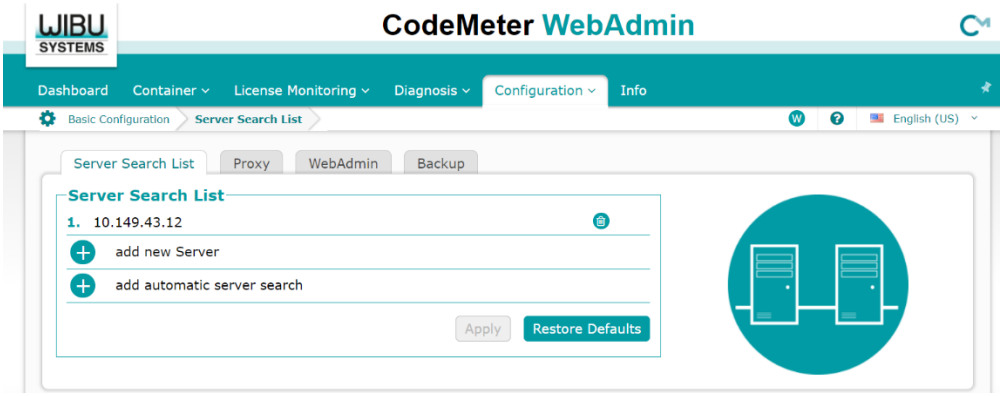


Figure 3-6 Add IP to Search List

4. The setup is complete. If the OPLS was completely set up, you can now test the connection by opening a tool that needs one of the licenses set on the OPLS.

3.1.3 Network Communication

The OPLS and the Client can only communicate with each other if there are no firewalls or other obstacles that prevent communication.

You can ping the IP address of the Server from the Client, to make sure communication is flowing. In addition, you need to unblock a specific Port.

Required Port

Make sure that Port 22350 is not blocked. The status you should be looking for is “listening”. This can checked by writing on the commandline `netstat -an`, which will show the status of different ports. See example below:

```
TCP    [::]:22350             [::]:0                 LISTENING
```

Figure 3-7 Port 22350 is Listening

Other potential status are:

- *connection refused* means that nothing is running on that port
- *accepted* means that something is running on that port

- *timeout* means that a firewall is blocking access

3.2 License Request Sequence

Licenses are requested in the following order:

- Local Licenses:
  - Device License
  - Named-User License
- Network-Based Licenses
  - Desktop NUNA
  - Desktop Floating
  - Server Edition: CI Team, CI Agent, CI MICROSAR based
- SpecCond Licenses
  - SpecCond Local
  - SpecCond NUNA
  - SpecCond Floating
  - SpecCond SE: CI Team, CI Agent, CI MICROSAR based

3.3 Sequence Restriction

The user can restrict the search of Licenses by specifying that he should only search either Desktop or Server licenses.

To make use of these restrictions, you need to set the environmental variable `VECTOR_LICENSE_EDITIONS` to either `DESKTOP`, `SERVER` or `ANY`. This environmental variable can be set for the Terminal, the specific user, or the computer.

3.3.1 Terminal

In case of the Terminal, the command would be the following with, for example, Powershell (if wanting to focus only on Server licenses):

```
Powershell: $env:VECTOR_LICENSE_EDITIONS="SERVER"
```

The environmental variable would be only valid through the opened terminal. If you want to use the GUI with this restriction, you would need to open the tool through the terminal (In this case, the GUI would´t find any license, since it is set to `SERVER`, and server licenses are not allowed to use the GUI). Once the terminal window is closed, the environmental

variable is reset. This means it would need to be set every time if you want to continue to use those restrictions.

3.3.2 System and User

Alternatively, you can set the environmental variable for your specific user or for the whole system (all users on the same computer). Restarting the computer will not reset this value, so it can be used if you wish to use those restrictions over the long term and would need to either be erased or set to `ANY` to go back to the normal sequence.

In this case, the environmental variable can be set through the Control Panel. You need to search for: “Edit the system environment variables”.

Set the variable `VECTOR_LICENSE_EDITIONS` on the Username or on System (you may need to create it) to either `DESKTOP`, `SERVER` or `ANY`.

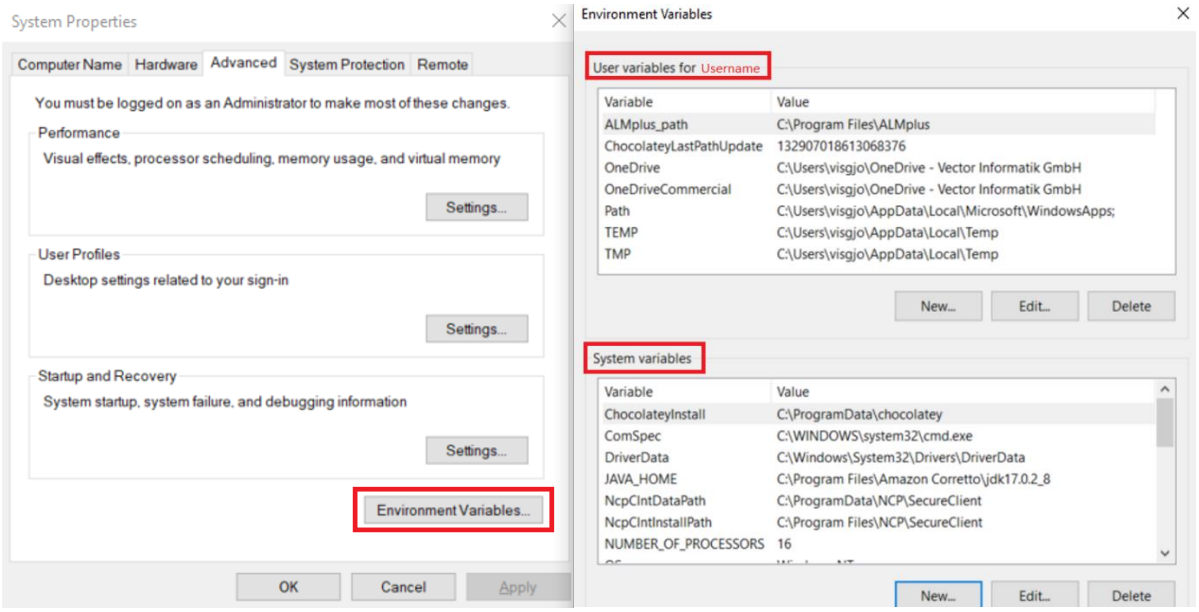


Figure 3-8 Setting the Syster or User Env. Variable

3.3.3 Priorization

As we saw, there are three ways to set the Variable. It can be set for the System, the User or the Terminal.

If set in more than one place, the following priorization is taken place.

1. System Env. Variable
2. User Env. Variable
3. Terminal Env. Variable

For example, even if the user and the terminal set their own variables, the System set variable would be taken as priority one, and its value is what would be used. For this reason, whenever you do not want any specific set up, it may be better to simply delete the variable, then to set it to ANY.

4 Legacy: FlexNet

4.1 FlexNet License Server Configuration

Flexnet Floating licenses can be provided by using a license server. The DaVinci Configurator application only accesses these licenses when:

- No DaVinci Configurator PRO License is available at the local PC (provided by a dongle or already locally installed)
- A valid license server connection is configured
- The `DaVinci_CFG` option within the settings.ini file is not explicitly set to ‘0’ (see also 4.1.2 and 4.1.3 for details)
- No installation warning is displayed in within the license dialog (Help > Licenses)

Otherwise the server isn’t accessed.

4.1.1 Server Configuration

The DaVinci Configurator application detects a license server connection if a `Settings.ini` file is either placed in the executable folder or in the common files folder for Vector applications. This file needs to be created manually.

The search order is identical to the order of appearance in table Table 4-1.

Setting File Locations	
Location	**Environment**
`DaVinciCFG.exe directory`	All
`%COMMONPROGRAMFILES(X86)%\Vector\DaVinci`	All
`DaVinciDEV.exe directory`	

Table 4-1 Settings File Locations



Note

If the `Settings.ini` file is placed on the “`%COMMONPROGRAMFILES(X86)%\Vector\DaVinci`” directory, it will be available for both DaVinci Configurator & Developer.

Placing it on the specific DaVinciCFG or DaVinciDEV directories, only makes it available for that tool.

Settings
The server based license model has its own set of settings:

> `Floating:` server based Flexnet licenses

For each license model an individual set of settings needs to be defined. Each consists of a defined number of optional server ‘option usage’ flags (described in 4.1.2) and the mandatory server connection configuration:

> `Host`: License server network ID

> `Port`: License server port

4.1.2 Server Option Usage Configuration

The usage of server license options which are available at the configured server can be defined for each option individually.

By default all options available at the server are used by the DaVinci Configurator application. To restrict this usage, the according section “`Floating`” of the settings file described in 4.1.1 may contain additional properties. One property is defined for each product option.

Server Option Usage		
Property	**Value**	**Description**
`DaVinci_CFG_DEV`	`{1|0}`	Allow or prevent locking a server based DaVinci Developer license to enable a RTE option within the DaVinci Configurator.
`DaVinci_CFG_MD`	`{1|0}`	Allow or prevent using a server based .MD product option.
`DaVinci_CFG_RTE`	`{1|0}`	Allow or prevent using a server based .RTE product option.
`DaVinci_CFG_WF`	`{1|0}`	Allow or prevent using a server based .WF product option.
`DaVinci_CFG`	`{1|0}`	Blocks access to all DaVinci Configurator server based licenses. The settings.ini file is used by the DaVinci Developer application too. With settings this flag to `0` the license server access of the DaVinci Configurator application is blocked. Use this option when a specific PC should not use DaVinci Configurator licenses but is allowed to use DaVinci Developer server based licenses.

Table 4-2 Server Options

4.1.3 Settings File Example

Following exemplary settings file content demonstrates the structure of the file used by the DaVinci Configurator application. Note that the ‘Server Option Usage’ flags are optional.

As a best practice measure, write and configure all the Options under the `Settings.ini` file. Keep in mind that the Host and Port values will come from your Server, therefore your IT Team.

> Floating license example:

```
[Floating]
Host=yourFloatingLicenseServer
Port=7777
DaVinci_CFG=1
DaVinci_CFG_DEV=0
DaVinci_CFG_MD=1
DaVinci_CFG_RTE=1
DaVinci_CFG_WF=0
```

4.1.4 DaVinci Developer License

A DaVinci Developer tool license (provided by a Vector NVLM “New Vector License Model” license or FlexNet license) enables the same tool functionality as a .RTE option, provided that both licenses – DaVinci Developer and DaVinci Configurator Pro – are located on the same PC.

The usage of a floating DaVinci Developer license by the DaVinci Configurator application can be prevented by limiting the usage of server based licenses (see 4.1.2 for details).

4.2 FlexNet Automation Server License (ASL) Handling

4.2.1 Installation and Delivery

FlexNet Automation Server Licenses are contained in .exe files which installs the ASL license once it is executed. Once the .exe is executed, you can confirm its correct installation by looking for the license at the Vector License Manager.

This license will be shared to you by the Vector Electronic Delivery Team.

4.2.2 License Use

The ASL license can be copied and pasted into several computers or Virtual Machines for its installation, becoming the only license that can be used multiple times.

4.2.3 Mixing NVLM and FlexNet Licenses

The first step of our Tool is to search for local NVLM licenses, if for example, the DaVinci Configurator License is not found (Options are irrelevant at this point), it continues searching for local FlexNet Licenses, and then for Floating FlexNet Licenses.

This means it is only possible to mix licenses when the Core DaVinci Configurator License is on FlexNet. For example, you could have a DaVinci Configurator Pro FlexNet Floating License mixed with NVLM Option WF and Option MD.



Note
As a tip, if you have a NVLM DaVinci Configurator License for an older Version (Example 5.20) and a newer DaVinci Configurator Version FlexNet Floating License (Example 5.22), you could deactivate the local NVLM license so the tool can then look for the Floating License. Any Options already locally activated through NVLM would also be detected.

5 Trouble Shooting

If you miss an activated DaVinci Configurator Classic license or option, please check the installation status within the ‘Licenses’ dialog first (Help > Licenses…).

Additionally, check if the required license is expired (Help > Licenses > Show Tool License Details).

More detailed information about tool licenses is displayed by following tools

- Vector License Client
  Shows details about Vector NVLM Licenses
- Vector License Manager
  Shows details about FlexNet based licenses.
- DaVinci Floating License Manager
  Shows details about DaVinci floating licenses (FlexNet based)

Those tools are provided by the external component setup. The latest version is available here: https://download.vector.com/davinci/ExternalComponentsSetup.zip.

Relevant for Flexnet-based licenses: If a DaVinci tool license is displayed in the Vector License Manager or Floating License Manager but not in the DaVinci Configurator’s license details dialog: please ensure that you have installed the component “DaVinci 64-bit Support” form the DaVinci External Components setup.

6 Abbreviations

6.1 Abbreviations

Abbreviation	Description
NVLM	New Vector License Model
LLM	Legacy License Model
OPLS	On-Premise License Server
SIP	Software Integration Package

7 Contact

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